A Multicenter Evaluation of the Iris iQ200 Automated Urine Microscopy Analyzer Body Fluids Module and Comparison With Hemacytometer Cell Counts

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Key Words: Cell counts; Serous fluid; Cerebrospinal fluid; Erythrocytes; Nucleated cells; Iris iQ200

DOI: 10.1309/WR1C5WNT6UFXNC6J

Abstract

Manual hemacytometer cell counting of body fluids is labor-intensive and requires skilled testing personnel. We performed a multicenter evaluation of the Iris iQ200 automated microscopy analyzer Body Fluids Module (Iris Diagnostics, Chatsworth, CA) and compared 350 iQ200 body fluid cell counts with manual hemacytometer cell counts. Within-run imprecision, expressed as coefficient of variation (CV), ranged from 2.6% to 5.9% for RBC counts between 875 and 475 × 10^6/L and from 4.2% to 6.5% for nucleated cell counts between 820 and 590 × 10^6/L. The lower limits of detection, based on a CV of 20% or less, were 30 and 35 × 10^6/L for RBCs and nucleated cells, respectively. There was very good agreement between automated iQ200 and manual body fluid cell counts based on slopes and r^2 values. The iQ200 has satisfactory performance for enumerating RBCs and nucleated cells in most body fluids, with the exception of cerebrospinal fluid specimens that contain low cell numbers.

Body fluid cell counts are a routine clinical laboratory test that provides valuable information for diagnosing and treating various medical conditions. Elevated WBC concentrations in cerebrospinal fluid (CSF) are observed in numerous conditions, including meningitis, encephalitis, and neurologic disorders, and in patients with involvement of the CSF by lymphoma and leukemia.[1] RBC counts are important for diagnosing intracerebral hemorrhage and for excluding a traumatic spinal tap as the cause of an elevated WBC count.[2] Cell counts on serous fluids collected by paracentesis are also useful for differentiating trauma, malignancy, and an infectious process.[3]

Body fluid cell counts have been traditionally performed by manually counting RBCs and nucleated cells under a microscope using hemacytometer counting chambers. Because it is difficult to correctly classify cells using counting chambers, stained cytocentrifuge preparations are recommended for differential cell counts.[4] Manual cell counting is extremely labor-intensive and requires highly experienced testing personnel, which is often in short supply when considering the need for 24-hour, 7-day-a-week availability. Manual cell counts also exhibit poor precision and have high interoperator variability.[5,6] One approach to overcome these inherent difficulties is to adapt automated hematology systems for enumeration of cells in various body fluids. The new hematology analyzers have combined electrical impedance with flow cytometry to more precisely and accurately count and differentiate between various cell types.[7] However, automated hematology analyzers can be hindered by electronic background noise that can lead to inaccurate cell counts with poor precision at cell numbers normally found in body fluids such as CSF.

We evaluated the analytic performance of the iQ200

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automated digital imaging system (Iris Diagnostics, Chatsworth, CA) for body fluid analysis and compared RBC and nucleated cell counts with manual hemacytometer chamber counts at 3 institutions.

Materials and Methods

Clinical Specimens

Body fluid cell counts were performed at 3 sites: (1) the Geffen School of Medicine at the University of California, Los Angeles; (2) the Medical University of South Carolina, Charleston; and (3) the M.D. Anderson Cancer Center, Houston, TX. Cell counts were determined on a total of 193 CSF and 157 serous fluid specimens. Only specimens received in the laboratory for routine analysis with sufficient volume for additional testing were used. The study was reviewed and approved by the human studies committees at each of the institutions. For CSF, 54.9%, 28.5%, and 16.6% of the specimens were analyzed at sites 1, 2, and 3, respectively. For serous fluid, 60.5% of the specimens were analyzed at site 1 and the remaining 39.5% were analyzed at site 2. There were 75 peritoneal fluid, 51 pleural fluid, 10 peritoneal lavage fluid, 4 peritoneal dialysate, 1 pericardial fluid, and 16 other fluid specimens described only as paracentesis fluid.

Precision Studies

For within-run imprecision, an aliquot of CSF with a very high RBC or WBC count was added to a pool of cell-free CSF or serous fluid to obtain body fluids with known cell concentrations. Multiple cell concentrations were prepared, and 10 aliquots at each cell concentration were counted on the same day using the iQ200 system. For between-run imprecision, level I and II iQ body fluid control samples (Iris Diagnostics) were assayed in duplicate for a period of 10 days. All precision studies were performed at site 1. The results are expressed as the CV percentage at different mean cell concentrations.

iQ200 Automated Cell Counting

The iQ200 Automated Urine Microscopy Analyzer (Iris Diagnostics) consists of 2 modules interlinked by cable communication and includes the iQ200 Automated Microscopy Module and the PC4 Analysis Processor/Results Processor. The iQ200 uses digital imaging and Auto Particle Recognition using neural network software to classify and quantitate particles in different categories. Samples do not require centrifugation. The images can be displayed on the screen for verification and manual editing. Up to 10,000 patient results and archived particle images can be stored in memory. Results for microscopic particles can be reported as particles per high-power field or per microliter. The iQ200 Body Fluids Module can be used on all iQ200 systems to examine and count RBCs and nucleated cells in body fluids. The device is indicated for analyzing CSF and peritoneal, pericardial, serous, and lavage fluids.

Two tubes were used for body fluid cell counts. One tube contained body fluid that was diluted with Iris Diluent. The other tube containing body fluid was diluted with iQ lysing reagent (Iris Diagnostics) to eliminate RBCs. Dilutions ranged from 5× to 100× depending on the cellularity of the specimen. Each sample was analyzed in duplicate (results were averaged) using the iQ200 automated digital imaging system with iQ200 Body Fluids Module software (Iris Diagnostics). The iQ200 requires a minimum of 1.25 mL of body fluid for testing using conical test tubes (Iris Diagnostics). CSF specimens were routinely diluted 5× so that only 0.25 mL was required for testing by the iQ200 system. Serous fluids were initially diluted 20× before testing, and a smaller (5×) or larger dilution was made depending on the cell counts. Nucleated cell counts were obtained for the lysed (prepared with Iris RBC lysing reagent) and unlysed (prepared with Iris Diluent) aliquots. RBC concentrations were calculated as the average of the 2 total cell counts for the unlysed fraction minus the average of the 2 nucleated cell counts for the lysed fraction. Calibration of the instrument was performed at the beginning of the study and at 1-month intervals. Before testing of body fluids each day, iQ focus and iQ body fluid control samples were run according to the manufacturer's instructions.

Manual Microscopic Cell Counting

Hemacytometer cell counts were performed on both iQ200 diluted samples after an additional 5× to 20× dilution. Two separate "improved" Neubauer hemacytometers (Bright-Line counting chambers, Fisher HealthCare, Houston, TX) were loaded, and both chambers of each hemacytometer were counted by bright-field microscopy using a 40× objective. Cells in the 4 small squares (0.5 × 0.5 mm) in each of the 4 corners on both sides of the hemacytometers were counted (0.2-µL volume per hemacytometer). If this area contained fewer than 100 RBCs or fewer than 100 nucleated cells, the large square (1 × 1 mm) in each of the 4 corners on both sides of the hemacytometers was counted for that particular cell type (0.8-µL volume per hemacytometer). For very low cell numbers (<100 cells per hemacytometer in the 8 large corner squares), an undiluted aliquot was counted. Results were expressed as the average of the 2 separate hemacytometer counts.

Statistical Analysis

EP Evaluator-CLIA software (David G. Rhoads Associates, Kennett Square, PA) was used for Deming regression analysis and supporting statistics.

Results

Analytic Performance of the iQ200

To determine linearity, the level II iQ body fluid con-

DOI: 10.1309/WR1C5WNT6UFXNC6J
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trol sample was diluted with iQ diluent to obtain varying concentrations of RBCs and nucleated cells. Each cell dilution was analyzed in triplicate. RBC and nucleated cell counts were found to be linear up to 52,500 and 2,850, respectively, based on mean ± SD linear regression analysis slopes of 1.00 ± 0.02. Within-run imprecision of the iQ200 was evaluated by analyzing CSF and serous fluid samples for RBCs and nucleated cells a total of 10 times within a single run. Imprecision ranged from 2.6% to 5.9% (CV) for RBC counts between 875 and 475 × 10^6/L and from 4.2% to 6.5% for nucleated cell counts between 820 and 590 × 10^6/L. Between-run imprecision during a 10-day period using iQ body fluid control samples was 7.6% and 4.6% for RBCs at mean cell concentrations of 24,586 and 49,766 and was 8.3% and 10.3% for nucleated cell concentrations of 1,675 and 2,787, respectively.

Imprecision was further examined at low cell concentrations to determine functional sensitivity. Overall, the imprecision increased (CV) with decreasing RBC and nucleated cell numbers, as illustrated in **Figure 1**. The lower limit of detection, based on a CV of 20% or less, was determined to be approximately 30 × 10^6/L for RBCs and approximately 35 × 10^6/L for nucleated cells.

Correlation Studies

Correlation data for CSF samples with RBC counts of 1 to 160,000 and nucleated cell counts of 1 to 1,000 by manual microscopy and the iQ200 are shown in **Figure 2**. Nucleated cell counts were determined in the presence of RBCs (Figure 2B) or after lysis of RBCs (Figure 2C). Deming regression analysis revealed slopes ranging from 0.96 to 1.05 that did not differ from 1.00 with 95% confidence **Table 1**. For RBC and nucleated cell counts after removal of RBCs, there was very good to excellent correlation based on r^2 values. On average, the iQ200 RBC counts were 0.1% higher and nucleated cell counts, after RBC lysis, were 3.9% lower than manual microscopy counts (Table 1). In contrast, nucleated cell counts in the presence of RBCs had slightly more scatter (Figure 2B), a weaker correlation (r^2, 0.62), and an average positive bias of 27.3%.

Correlation data for serous fluids with RBC counts of 1 to 160,000 and nucleated cell counts of 1 to 4,000 are shown in **Figure 3**. The slope for RBC counts was 0.90 and statistically different from 1.00. In contrast, the slopes for nucleated cell counts with and without RBC lysis did not differ from 1.00 (Table 1). There was excellent correlation for RBC and nucleated cell counts after RBC lysis based on r^2 values of 0.94 or more. On average, iQ200 RBC and nucleated cell counts after RBC lysis were 11.1% and 2.2%, respectively, lower than manual cell counts. Nucleated cell counts in the presence of RBCs displayed considerably more scatter (Figure 3B), a weak correlation, and an average positive bias of 18.7% (Table 1).

Clinical Performance of the iQ200

The ability of the iQ200 to correctly classify CSF specimens with normal and abnormal RBC and nucleated cell counts was examined using manual hemacytometer counts as the comparison method. As shown in **Table 2**, the agreement rate was 93% or more for normal and abnormal RBC counts. The agreement rate was not as good for nucleated cells and was approximately 70% when examining samples with 5 × 10^6 or fewer cells per liter and probably reflects the greater imprecision for nucleated cell counts less than the selected cutoff values.

Discussion

Iris Diagnostics has developed a software module that

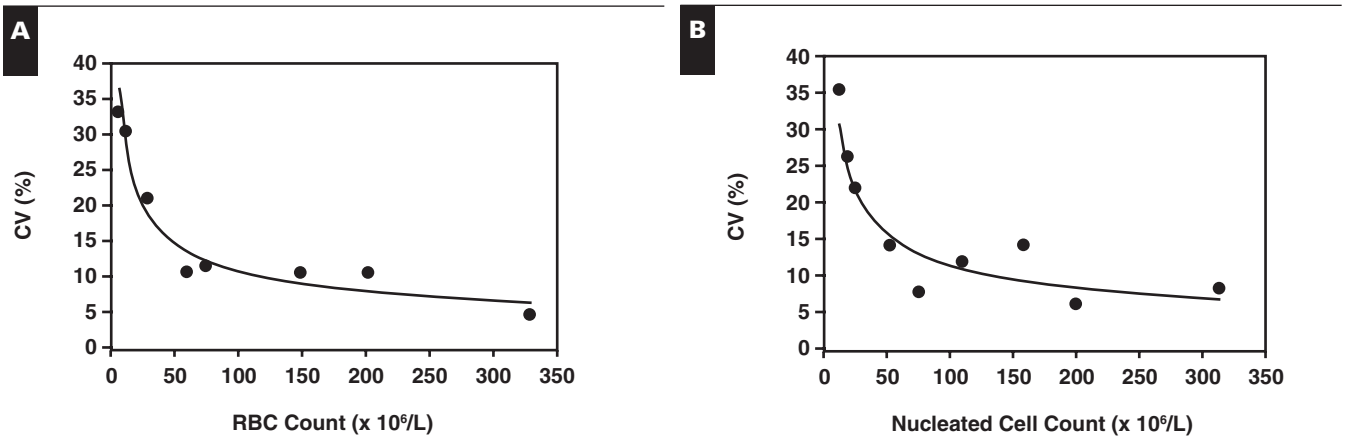

Figure 1 Within-run imprecision of the iQ200 at low body fluid cell concentrations. Cerebrospinal fluid (CSF) specimens containing RBCs (**A**) or nucleated cells (**B**) were diluted with a cell-free CSF and analyzed 10 times within a day. CV, coefficient of variation.

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DOI: 10.1309/WR1C5WNT6UFXNC6J

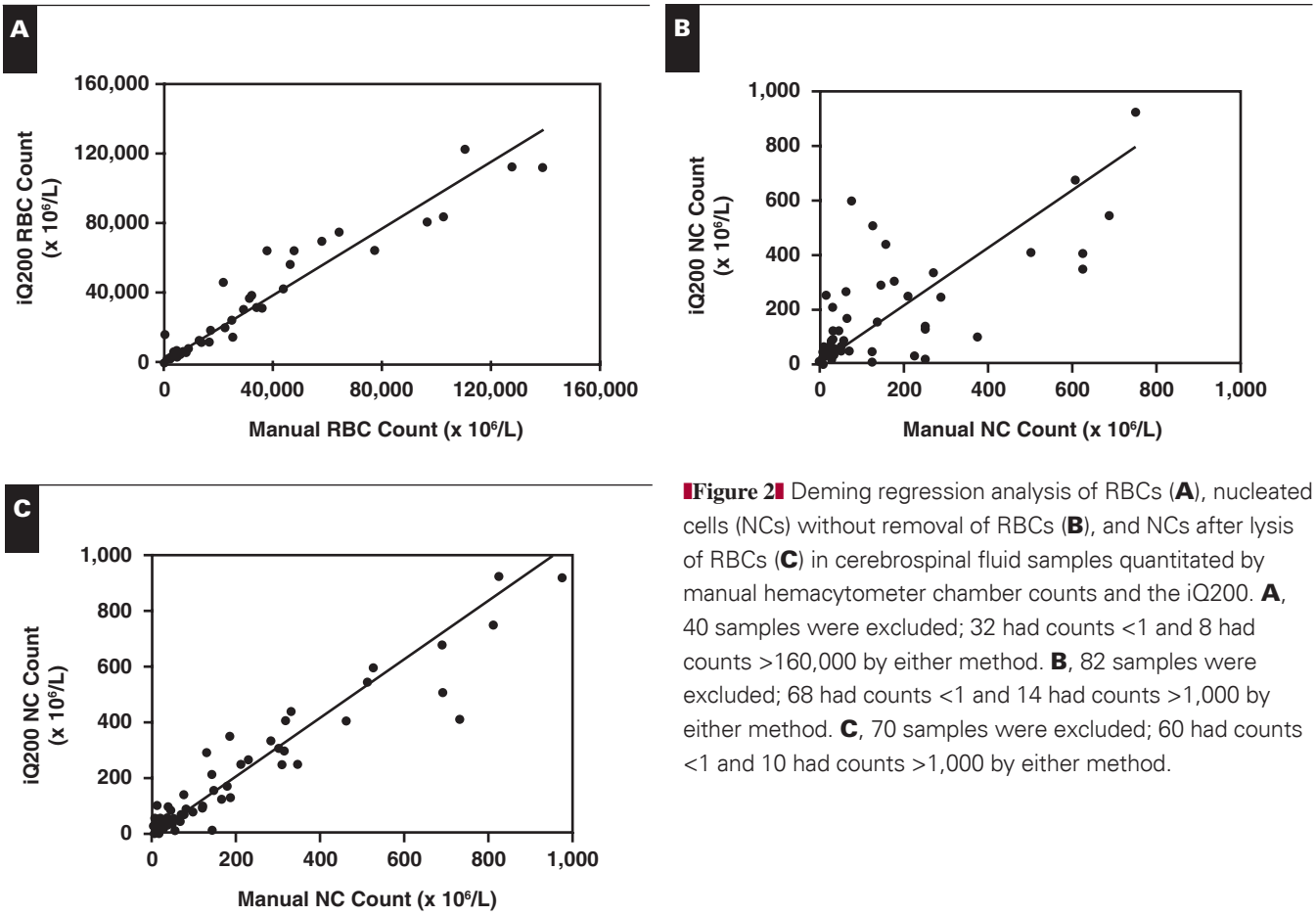

Figure 2 Deming regression analysis of RBCs (**A**), nucleated cells (NCs) without removal of RBCs (**B**), and NCs after lysis of RBCs (**C**) in cerebrospinal fluid samples quantitated by manual hemacytometer chamber counts and the iQ200. **A**, 40 samples were excluded; 32 had counts <1 and 8 had counts >160,000 by either method. **B**, 82 samples were excluded; 68 had counts <1 and 14 had counts >1,000 by either method. **C**, 70 samples were excluded; 60 had counts <1 and 10 had counts >1,000 by either method.

Table 1
Deming Regression Analysis of RBC and Nucleated Cell Counts in Body Fluids Quantitated by the iQ200 and Manual Hemacytometer Chamber Counting

	No. of Samples	Slope (95% CI)	Intercept (95% CI)	r^2	Mean % Difference
CSF*					
RBCs	153	0.96 (0.92 to 1.00)	393 (–510 to 1,296)	0.95	0.1
NCs, unlysed	111	1.05 (0.90 to 1.21)	7 (–20 to 33)	0.62	27.3
NCs, lysed	123	1.04 (0.96 to 1.12)	3 (–15 to 20)	0.84	–3.9
Serous fluid*					
RBCs	130	0.90 (0.86 to 0.94)	–215 (–1,864 to 1,433)	0.94	–11.1
NCs, unlysed	132	0.96 (0.80 to 1.12)	133 (–51 to 317)	0.53	18.7
NCs, lysed	142	0.98 (0.94 to 1.02)	3 (–45 to 50)	0.95	–2.2

CSF, cerebrospinal fluid; NCs, nucleated cells.
* The cells in fluids were counted in the presence of RBCs (unlysed) or after removal of RBCs by treatment with lysis reagent (lysed).

can be used with the iQ200 automated urine microscopy analyzer to count RBCs and nucleated cells in body fluid specimens. The Body Fluids Module separates particles into 2 categories, RBCs and nucleated cells, based on size. The individual images are stored and can be displayed on the screen for verification and/or manual editing. Body fluid RBC and nucleated cell counts require a total of about 5 minutes to perform (including dividing specimens into aliquots and performing dilutions) using the iQ200, which is considerably faster and less labor-intensive than manual hemacytometer cell counts.

A major limitation of all automated cell counters is poor precision at low cell concentrations. This limitation is especially problematic for CSF specimens because they frequently have nucleated and WBC counts less than 50×10^6/L. Our results indicate that the iQ200 has a lower detection limit of 30×10^6/L for RBCs and 35×10^6/L for nucleated cells, based on within-run CVs of 20% or less. This limit is similar to the detection limits of 25×10^6/L for RBCs and 18×10^6/L for WBCs that we determined for the iQ200 when testing urine samples.[8] The

DOI: 10.1309/WR1C5WNT6UFXNC6J
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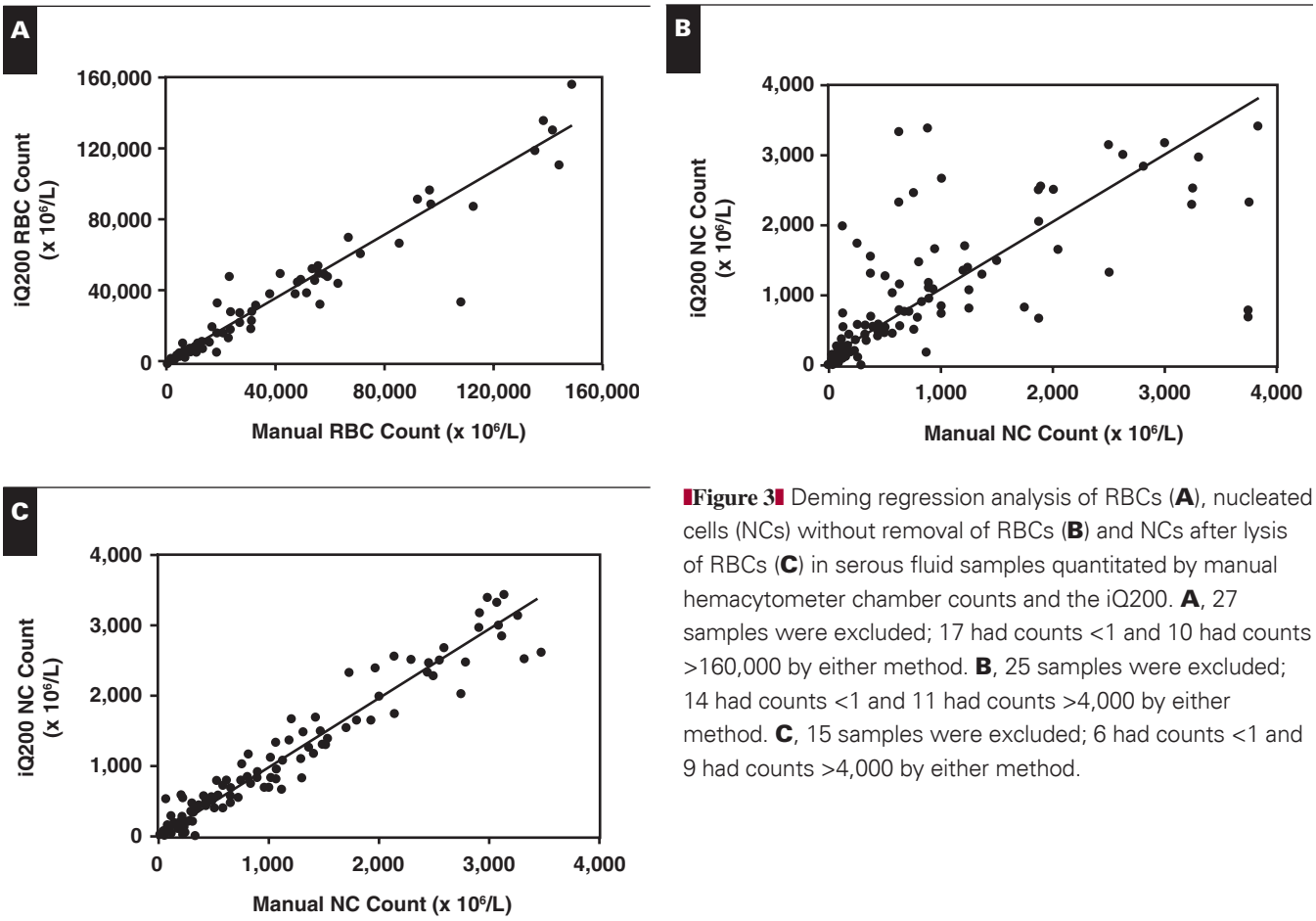

Figure 3 Deming regression analysis of RBCs (**A**), nucleated cells (NCs) without removal of RBCs (**B**) and NCs after lysis of RBCs (**C**) in serous fluid samples quantitated by manual hemacytometer chamber counts and the iQ200. **A**, 27 samples were excluded; 17 had counts <1 and 10 had counts >160,000 by either method. **B**, 25 samples were excluded; 14 had counts <1 and 11 had counts >4,000 by either method. **C**, 15 samples were excluded; 6 had counts <1 and 9 had counts >4,000 by either method.

Table 2
Agreement Between iQ200 and Manual Cell Counts Based on Cerebrospinal Fluid Cutoff Values

		Agreement*	
Cell Type	**Upper Limit of Normal (× 10^6/L)**	**Normal**	**Abnormal**
RBCs	10	43/46 (93)	139/147 (94.6)
Nucleated cells	5	61/85 (72)	89/108 (82.4)

* Results were considered normal or abnormal based on manual hemacytometer cell counts. For nucleated cell manual counts, the RBCs were removed by treatment with lysis reagent before counting. Data are given as number/total (percentage).

iQ200 detection limits are adequate for many biologic fluids, such as peritoneal and pericardial fluids, because they typically have WBC clinical decision cutoffs of 500 and 1,000 × 10^6/L for peritonitis and pericarditis, respectively.[9] Unfortunately, the precision of the iQ200 for nucleated cells may not be adequate for counting all CSF specimens because WBC counts of more than 5 × 10^6/L and more than 30 × 10^6/L are considered abnormal in adults and newborns, respectively.[10] This is the most likely explanation for the modest overall normal-abnormal rate of approximately 77% that we observed for CSF nucleated cell counts obtained by the iQ200 when using manual hemacytometer counts as the comparison method (Table 2). However, it is possible to screen CSF samples with the faster and less labor-intensive iQ200 Body Fluids Module so that only samples with low nucleated cell counts require reflex testing using manual hemacytometer counts.

Other automated cell counters also have poor precision at low WBC concentrations. For example, the Sysmex XE-2100 automated hematology analyzer has been shown to have a lower detection limit of 50 × 10^6/L for WBCs.[11,12] A similar

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DOI: 10.1309/WR1C5WNT6UFXNC6J

lower detection limit of 50×10^6/L for WBCs was also reported using the Bayer Advia 120.[11,13] For CSF specimens with normal WBC counts of 2×10^6/L, the Bayer Advia 120 was shown to have unacceptably high CVs of 30% or more.[14,15] Finally, the Abbott CellDyn-4000 was reported to have CVs of 30% and 23% at WBC concentrations of 5 and 10×10^6/L, respectively.[16] These data indicate that none of the currently available automated cell counters have adequate precision for reliable WBC counts in CSF specimens. This finding also holds true for manual hemacytometer cell counts because they have been shown to be even more imprecise than automated counting systems at low cell concentrations.[17]

There was very good agreement between the automated iQ200 and manual RBC and nucleated cell counts for body fluid specimens across 3 testing sites. The slopes based on Deming regression analysis were not statistically different from 1.00, except for RBC counts in serous fluid (Table 1). In addition, the iQ200 produced lower RBC counts in serous fluid specimens that, on average, were 11.1% lower than manual cell counts. The reason for lower RBC counts in serous fluid by the iQ200 is unclear but could reflect inaccuracies associated with the manual cell counting method because higher cell counts were more common when examining serous fluid specimens (Figures 2 and 3). It is interesting to note that removal of RBCs had a dramatic effect on nucleated cell counts and reduced the amount of scatter (based on r^2) around the regression line. In a previous study, we also found good agreement between the iQ200 and manual chamber counts for RBCs and WBCs in urine samples.[8] Similar to our findings, other investigators have demonstrated good correlation when comparing body fluid cell counts by automated hematology analyzers such as the Sysmex XE-2100 (r^2 values of 0.90-0.98), the Bayer Advia 120 (r^2 value of 0.92 for WBCs), and the Abbott CellDyn-4000 (r^2 value of 0.98 for WBCs) with manual chamber counts.[11-13,16]

The iQ200 automated urine microscopy analyzer with the body fluids software module has acceptable analytic performance for enumerating RBCs and nucleated cells in most body fluids, with the exception of CSF specimens with low cell concentrations. Major advantages of this system include the excellent precision at high cell concentrations, good agreement with standard hemacytometer counts, and a significant reduction in the time required to perform cell counts without sacrificing the visual benefit of microscopic analysis. The drawbacks include poor precision at low cell numbers, making it unsuitable as the sole method for CSF cell counting, and the continued need to manually perform differential cell counts on stained cytocentrifuged preparations because the system cannot distinguish between nucleated cell types.

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Supported by Iris Diagnostics.

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Acknowledgments: We thank Elizabeth Clayton, Children's Hospital at the Medical University of South Carolina, for providing some of the body fluid data that were included in the study.

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DOI: 10.1309/WR1C5WNT6UFXNC6J
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